

TEMUCO, Chile

WMO: 857430

Lat: 38.768S

Lon: 72.632W

Elev: 92

StdP: 100.22

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.9	-0.8	-3.3	2.9	1.3	-2.0	3.2	0.8	9.0	10.4	7.8	10.0	0.8	90	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.3	28.6	18.5	26.3	17.6	24.7	16.8	19.1	27.0	18.2	25.2	17.3	23.6	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.6	21.7	15.2	10.9	20.6	14.6	10.5	19.8	54.7	27.1	51.7	25.3	49.0	23.5	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.3	5.5	-4.2	34.5	1.2	2.3	-5.1	36.1	-5.8	37.4	-6.4	38.7	-7.3	40.4		
			-4.4	21.2	1.1	1.1	-5.2	22.0	-5.8	22.6	-6.5	23.3	-7.3	24.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	16.6	16.9	14.9	11.8	9.8	8.3	7.3	8.3	9.6	11.4	12.9	15.0
	DBStd	4.09	2.46	2.71	2.47	2.72	2.75	2.85	2.64	2.50	2.32	2.15	2.22	2.47
	HDD10.0	320	0	0	1	13	37	66	90	66	35	10	3	1
	HDD18.3	2391	64	54	111	197	264	301	342	312	263	216	162	104
	CDD10.0	1002	206	194	151	66	32	14	6	12	22	52	91	157
	CDD18.3	32	11	15	3	0	0	0	0	0	0	0	0	3
	CDH23.3	722	225	286	92	11	0	0	0	0	2	6	12	90
(14) Wind	WSAvg	2.2	2.5	2.3	1.9	1.7	1.9	2.3	2.3	2.4	2.1	2.2	2.4	2.5
	PrecAvg	1418	34	37	58	102	212	249	205	180	117	91	73	55
	PrecMax	1761	96	142	151	208	434	517	313	349	196	293	173	147
	PrecMin	872	5	1	18	28	68	64	58	91	52	14	30	1
	PrecStd	242	25	30	29	49	94	94	67	57	43	59	35	37
	DB	32.2	34.3	29.1	24.9	19.6	16.0	16.1	18.0	22.1	23.5	24.9	28.7	
	MCWB	19.7	18.7	18.7	16.1	14.2	12.9	12.1	11.8	13.1	15.4	16.7	18.5	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	27.8	29.4	25.8	21.8	17.2	14.6	13.9	15.5	18.3	20.1	22.0	25.8	
	2% MCWB	18.4	18.7	17.0	15.2	13.8	12.1	11.4	11.2	12.3	14.4	16.0	17.5	
	5% DB	25.7	26.8	23.6	19.3	15.8	13.6	12.8	14.1	16.4	18.2	20.1	23.4	
	5% MCWB	17.5	17.8	16.2	14.3	13.1	11.5	10.6	10.7	11.7	13.2	14.8	16.7	
	10% DB	23.8	24.2	21.5	17.4	14.4	12.7	11.9	12.9	14.9	16.6	18.7	21.6	
	10% MCWB	16.7	16.6	15.2	13.6	12.1	10.9	9.9	10.1	10.9	12.3	13.9	15.7	
	WB	20.5	20.4	19.4	17.3	15.6	13.7	13.2	13.5	14.6	16.5	18.0	19.5	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	29.2	31.0	26.8	22.1	18.0	15.1	14.6	16.2	18.6	21.9	23.0	26.8	
	2% WB	18.9	19.1	17.8	16.0	14.3	12.6	11.8	12.2	13.2	14.8	16.2	18.1	
	2% MCDB	26.6	27.5	24.2	20.1	16.4	14.0	13.3	14.5	17.0	19.5	20.9	24.5	
	5% WB	18.0	18.1	16.7	14.9	13.3	11.7	11.0	11.3	12.2	13.6	15.2	17.0	
	5% MCDB	24.9	25.6	22.1	18.5	15.1	13.2	12.5	13.3	15.6	17.5	19.5	22.8	
	10% WB	16.9	17.0	15.7	14.0	12.5	11.1	10.1	10.5	11.2	12.6	14.2	15.9	
	10% MCDB	23.1	23.6	20.4	16.9	14.2	12.5	11.5	12.4	14.3	16.0	17.9	20.8	
(35) Mean Daily Temperature Range	MDBR	14.8	15.3	14.1	11.7	8.6	7.0	7.7	8.8	10.6	10.9	11.5	13.1	
	5% DB	18.4	19.8	17.9	15.2	10.2	7.3	8.7	10.9	14.0	14.1	14.5	16.5	
	5% WB	9.6	9.8	9.7	9.1	7.0	5.2	6.2	7.1	8.6	8.7	8.9	9.2	
	MCDBR	16.6	17.8	15.8	13.1	8.3	6.2	6.9	8.4	12.2	12.9	13.4	15.2	
	MCWBR	9.3	9.1	9.0	8.2	5.9	4.6	5.2	5.8	7.8	8.2	8.6	8.9	
(40) Clear-Sky Solar Irradiance	taub	0.325	0.341	0.333	0.334	0.335	0.318	0.312	0.322	0.332	0.328	0.325	0.330	
	taud	2.530	2.475	2.493	2.472	2.467	2.536	2.548	2.506	2.447	2.478	2.495	2.497	
	Ebn at Noon	1007	966	931	862	792	782	815	865	917	969	1000	1006	
	Edh at Noon	109	110	99	88	77	67	70	84	102	109	112	114	
(44) All-Sky Solar Radiation	RadAvg	7.79	6.76	5.07	3.23	1.98	1.44	1.73	2.43	3.88	5.16	6.53	7.49	
	RadStd	0.39	0.34	0.33	0.22	0.22	0.20	0.20	0.27	0.34	0.46	0.47	0.62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page